

Q1. **Scenario:** A sports team tracks player performance across 12 matches.
Task: Create a (12, 5) array with random integers (0 to 100, representing scores). Mask the values where the player's score is less than the team's average score for that match. Replace these values with -1.

Expected Output Example: Array with some values replaced by -1.

Q2. **Scenario:** A business tracks monthly revenue and expenses for 3 departments across 6 months.

Task: Create two (3, 6) random matrices for revenue and expenses. Perform element-wise multiplication and sum along the row axis to calculate the total profit for each department.

Expected Output Example: A 1D array with the total profit per department

Q3. **Scenario:** A company tracks salaries of 15 employees in 3 departments.

Task: Create a (3, 15) matrix of random salaries. Identify and print the department(s) where all employees have salaries above ₹50,000. Replace salaries less than ₹30,000 with the department's average salary.

Expected Output Example: Original Salary matrix and Updated salary matrix.

Q4. Calculate cumulative product of values, ignoring zeroes:

Test case 1:

Input: Series([1, 2, 0, 4, 5])

Output: Series([1, 2, 2, 8, 40])

Q5. Write a python program which Shift elements of a Series and fill shifted positions with custom values:

Test case 1:

Input: Series([10, 20, 30, 40]), shift: 2, fill_value: -1

Output: Series([-1, -1, 10, 20])

Q6. Generate random numbers in a Series and count how many are greater than the standard deviation:

Test case 1:

Input: Series([1.2, 2.5, 3.6, 4.1, 5.9])

Output: 2

Q7. Convert negative values in a Series to absolute values:

Test case 1:

Input: Series([-10, 5, -3, 2, -7])

Output: Series([10, 5, 3, 2, 7])

Q8. Find positions of prime numbers in a Series and return the count:

Test case 1:

Input: Series([10, 7, 3, 15, 17])

Output: 3